

Chapter → 3

Strings

- String is a datatype in Python.
- String is a sequence of characters enclosed in Quotes.

- We can Primarily write a String in these three ways

a = 'harry' # Single Quoted String

b = "Mahip" # Double Quoted String

c = ''' Mohit ''' # Triple Quoted String

★ String Slicing

A String in Python can be sliced for getting a part of String.

Consider the following String:

Name = "Harry" ⇒ length = 5

↓ ↓ ↓ ↓ ↓
 0 1 2 3 4
 (-5) (-4) (-3) (-2) (-1)

The index in a String starts from 0 to (length-1) in Python. In order to slice a String, we use the following Syntax:

Sl = Name[ind-start : ind-end]

first index included

last index is not included

Sl[0:3] returns "har" → characters from 0 to 3

Sl[1:3] returns "ar" → characters from 1 to 3

★ Slicing with skip value

We can provide a skip value as a part of our slice like this:

```
word = "amazing" # "mzn"  
word[1:6:2]
```

• Other Advanced slicing techniques:

```
word = "amazing"  
word[:7] # word[0:7] amazing  
word[0:] # word[0:7] amazing
```

★ String Functions

1) len() function - This function returns the length of the strings.

```
len("Malip") # 5
```

2) String.endswith("hip") - This function tells that whether the variable string ends with the string "hip" or not. If string is "Malip", it returns true for "hip" since Malip ends with hip.

```
MyString = "Malip"
```

~~is~~ MyString.endswith("hip")

3) String.count("c") - counts the total number of occurrences of any character.

```
String = "abracadabra"
```

```
count = String.count("c")
```

```
print(count) # output 1
```

4) String.capitalize() - This function capitalizes the only first character of a given string.

```
s = "hello world"
```

```
capitalized_string = s.capitalize()
```

```
print(capitalised_string)
```

5) string.find(word) → This function finds a word and returns the index of first occurrence of that word in the string

```
s = "hello world"
```

```
index = s.find("world")
```

```
print(index)
```

output : 6

6) string.replace(old word, new word) → This function replace the old word with new word in the entire string.

```
s = "hello world"
```

```
replaced-string = s.replace("world", "python")
```

```
print(replaced-string)
```

There are many functions like this in strings.

★ Escaping Sequence Characters

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Practice Set

1) Write a python program to display a user entered name followed by Good Afternoon using input function.

2) Write a program to fill in a letter template given below with name and date.

```
letter = '''
```

```
Dear <|Name|>,
```

```
you are selected!
```

```
<|Date|>
```

```
'''
```

3) Write a program to detect double space in a string.

4) Replace the double space from Problem 3 with single space.

5) Write a program to format the following letter using

Escape sequence characters.

```
letter = "Dear Harry, this python course is nice. Thanks"
```